Customer Shopping Behavior Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 customer purchases across multiple product categories. The objective is to understand purchasing patterns, customer demographics, subscription behavior, and product performance to support data-driven business decisions.

The project follows an end-to-end analytics workflow involving data cleaning and exploratory data analysis in Python, business analysis using PostgreSQL and SQL, and interactive dashboard creation using Power BI.

---

2. Dataset Summary

- Rows: 3,900

- Columns: 18

- Key Features:

    - Customer demographics (Age, Gender, Location, Subscription Status)

    - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)

    - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

    - Missing Data: 37 values in Review Rating column

---

3. Exploratory Data Analysis using Python

The dataset was first processed and cleaned using Python and Pandas.

Data Cleaning

- Imported the dataset using **Pandas**.
- Performed initial exploration using info() and describe().
- Checked for missing values and data inconsistencies.
- Imputed missing Review Rating values using category-wise median ratings.
- Standardized column names using **snake_case** naming convention.

- Verified duplicate records and ensured data consistency.

Feature Engineering

- Created **age_group** for customer segmentation.
- Converted purchase frequency categories into numerical **purchase_frequency_days.**
- Prepared the dataset for SQL-based business analysis.
- Loaded the cleaned dataset into **PostgreSQL** using **SQLAlchemy**.

---

4. Exploratory Data Analysis (EDA)

Key Statistics

- Total Revenue Generated: $233,081
- Average Purchase Amount: $59.76
- Average Review Rating: 3.75
- Total Customers Analyzed: 3,900

Revenue Analysis

Category-wise revenue analysis revealed that Clothing generated the highest revenue, followed by Accessories, Footwear, and Outerwear.

Category	Revenue
Clothing	$104,264
Accessories	$74,200
Footwear	$36,093
Outerwear	$18,524

Demographic Analysis

Young Adult customers contributed the highest revenue among all age groups. Spending behavior remained relatively consistent across age categories.

Subscription Analysis

Approximately 27% of customers were subscribers while 73% were non-subscribers. Average spending between subscribers and non-subscribers was found to be nearly identical.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

---

5. Business Analysis using SQL

PostgreSQL was used to answer key business questions and derive actionable insights.

Revenue by Gender

Compared total revenue generated by male and female customers.

Result:

- Male Revenue: $157,890
- Female Revenue: $75,191

	gender text	revenue numeric
1	Female	75191
2	Male	157890

High-Spending Discount Users

Identified customers who used discounts but still spent above the overall average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Total rows: 839 Query complete 00:00:(

Top Products by Rating

Determined products with the highest average review ratings to identify customer favorites.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Subscription Analysis

Compared average spending and total revenue between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Customer Segmentation

Classified customers into segments using previous purchase history:

- New Customers
- Returning Customers
- Loyal Customers

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Top Products per Category

Used SQL Window Functions and ROW_NUMBER() to identify the top 3 most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Revenue by Age Group

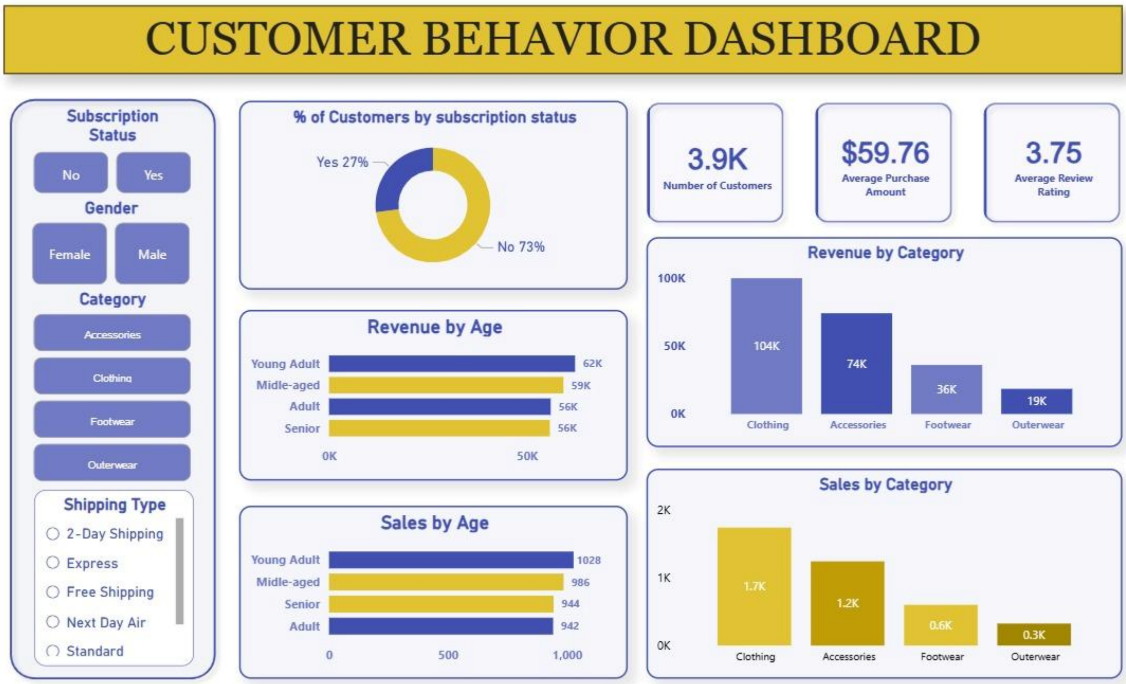
Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

---

6. Power BI Dashboard

An interactive dashboard was developed using Power BI to visualize customer behavior and business performance metrics.



7. Key Insights

1. Clothing is the highest revenue-generating category, contributing over $104K in revenue.
2. Young Adult customers generate the highest revenue among all customer age segments.
3. Non-subscribers account for approximately 73% of customers, indicating opportunities to increase subscription adoption.
4. Subscription status has minimal impact on average spending behavior.

5. Accessories and Clothing consistently outperform other categories in both revenue and sales volume.
6. Customer purchasing behavior remains relatively stable across age groups.

---

8. Business Recommendations

Increase Subscription Adoption

Introduce exclusive discounts, loyalty rewards, and personalized offers to encourage more customers to subscribe.

Focus on High-Revenue Categories

Allocate more marketing resources to Clothing and Accessories, which contribute the majority of revenue.

Target Young Adult Customers

Develop targeted campaigns aimed at the highest revenue generating age segment.

Strengthen Customer Retention

Implement loyalty programs to convert repeat buyers into long-term customers.

Promote Best-Selling Products

Highlight top-performing products through personalized recommendations and marketing campaigns.

---

9. Conclusion

This project demonstrates how customer shopping data can be transformed into actionable business insights through data analytics. By combining Python, PostgreSQL, SQL, and Power BI, meaningful patterns in customer demographics, purchasing behavior, and product performance were identified.

The resulting dashboard provides an interactive platform for monitoring business performance and supporting strategic decision-making. The insights derived from this analysis can help businesses improve customer engagement, optimize marketing efforts, and drive sustainable revenue growth.